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Aircraft propulsion system with voltage regulator

Abstract

Aspects of the disclosure generally relate to an aircraft propulsion system for an aircraft. The aircraft propulsion system can include at least an electric power source, an electric machine, and a voltage regulator. The voltage regulator regulates the electrical power provided to the electric machine from the electrical power source. The electric power source is capable of providing an AC or DC electrical output and can include a combustion engine with a generator or an electrical storage device.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION(S)

(1) This application claims priority to and benefit of U.S. patent application Ser. No. 16/688,108, filed Nov. 19, 2019, now U.S. Pat. No. 11,440,673, issued Sep. 13, 2022, which is incorporated herein in its entirety.

TECHNICAL FIELD

(2) This disclosure generally relates to electrical aircraft propulsion systems and more specifically relates to the controlling of voltage for such systems.

BACKGROUND

(3) Aircraft with electrical propulsion systems are not currently commonplace, although more interest is being shown. Electrical propulsion systems face unique challenges as compared to combustion propulsion systems, especially if aircraft with electrical power systems are to operate safely in the same or similar flight envelope of aircraft with combustion power systems.

(4) One particular technical issue that must be overcome is that electrical propulsion systems include electrical conductors that must be insulated so that throughout the flight envelope the electrical conductor does not experience a partial discharge. This is especially true for aircraft having a flight envelope including higher elevations where the reduced density of the air provides less electrical insulation, which, all other things being equal, increases the possibility of a partial discharge.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings:

(2) FIG. 1 is a schematic illustration of an aircraft with an aircraft propulsion system.

(3) FIG. 2 is a schematic illustration of a variation of the aircraft with the aircraft propulsion system of FIG. 1.

(4) FIG. 3 is a schematic illustration of another variation of the aircraft with the aircraft propulsion system of FIG. 1.

(5) FIG. 4 is a schematic illustration of a variation of the aircraft with the aircraft propulsion system of FIG. 3.

(6) FIG. 5 is a flow chart diagram illustrating a method of operating an electric distribution system that can be carried out at least in part by the aircraft with an aircraft propulsion system of FIG. 1-FIG. 4.

DETAILED DESCRIPTION

(7) Aspects of the present disclosure describe an aircraft propulsion system with an electric machine that rotates at least one propeller, fan, turbine, or similar device. As illustrated by way of non-limiting example, the at least one propeller can be any propulsor or propelling device. Electric power is provided to the electric machine from an electric power source via electrical conductors. A voltage regulator can regulate the voltage provided to the electrical machine from the electrical power source and therefore the voltage provided to at least a portion of the electrical conductors. Optionally, the voltage regulator can regulate the voltage provided to one or more inverters, where

the one or more inverters provide voltage to the electric machine. The voltage of the supplied power can be a function of the ambient air pressure of the electrical conductors. In cases where the electrical conductor is exposed to atmospheric pressure, the ambient air pressure is a function of the altitude of the aircraft, making the voltage a function of altitude.

(8) Connection references (e.g., attached, coupled, connected, secured, fastened, and joined) are to be construed broadly and can include intermediate members between a collection of elements and relative movement between elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and in fixed relation to one another. The exemplary drawings are for purposes of illustration only and the dimensions, positions, order and relative sizes reflected in the drawings attached hereto can vary.

(9) FIG. 1 is a schematic illustration of an aircraft **10** with an aircraft propulsion system **12**. The aircraft propulsion system **12** includes at least one electric power source **14**, at least one electric machine **16**, at least one voltage regulator **18**, and at least one propeller **20**. The propeller **20** is operably rotated by the electric machine **16**. The electric machine **16** is powered by the electric power source **14** via the voltage regulator **18**. The voltage regulator **18** regulates the electrical power provided to the electric machine **16** from the electrical power source **14**.

(10) The electric power source **14** can be any configuration of elements that provides an electrical output, such as alternating current (AC) or direct current (DC). The electric power source **14** can be an electrical storage device. Non-limiting examples of the electrical storage machine can include one or more of a battery, a capacitor, or a thermal system. Additionally or alternatively, the electric power source **14** can be a combustion engine with a generator or rectifier. By way of non-limiting example, the combustion engine can be a turbine engine or piston engine.

(11) The electric machine **16** can be any electromagnetic machine that can transform electrical energy to mechanical energy or mechanical energy to electrical energy. That is, the electric machine **16** includes one or more stator/rotor combinations. Non-limiting examples include an electric generator, an electric motor, or a combination electric generator/motor. The electric generator or electric motor can be an AC or DC generator or motor.

(12) The voltage regulator **18** can include any number of electrical or mechanical components resulting in an output from the voltage regulator **18** that is a stable, predetermined, and controlled value. The output value from the voltage regulator **18** can be greater than the input voltage to the voltage regulator **18**, although it is contemplated that the output value can be equal to or less than input value. By way of non-limiting example, the voltage regulator **18** can be located within or directly coupled to the electrical power source **14**. Alternatively, the voltage regulator **18** can include any combination of one or more of an inverter, a converter, a rectifier, a capacitor, a resistor, a diode, an inductor, a transistor, a regulator, a switch, a transformer, or an amplifier. The voltage regulator **18** can function as a power sink or a power source, where the voltage regulator **18** can be used to control power, voltage, or current.

(13) A first electrical conductor **22** can connect the electric power source **14** to the voltage regulator **18**. A second electrical conductor **24** can connect the voltage regulator **18** to the electric machine **16**. At least a portion of the first electrical conductor **22** or the second electrical conductor **24** is located in at least a portion of the aircraft **10**.

(14) The first electrical conductor **22** and the second electrical conductor **24** can include electrical insulation. The electrical insulation can be any known electrical insulation material used to reduce the likelihood of partial discharge, aging of electrical conductors, or other type of insulation failure. By way of non-limiting example, a first electrical insulation **32** can surround, wrap, coat, or otherwise insulate the first electrical conductor **22**. Similarly, a second electrical insulation **34** can surround, wrap, coat, or otherwise insulate the second electrical conductor **24**. The first electrical insulation **32** can have a different composition or different dimensions than the second electrical insulation **34**. Dimensions or size can include, but is not limited to, a gauge, a diameter, a cross-sectional area, or other measurable aspect of the insulation.

(15) At least a portion of the first electrical conductor **22** with the first electrical insulation **32** or the second electrical conductor **24** with the second electrical insulation **34** can be exposed to the pressure of the surrounding environment, and will be referred to as ambient pressure. It is contemplated that in most cases, the ambient pressure will be the environmental pressure outside the aircraft for the current altitude, which is the atmospheric pressure. In such an instance, the ambient pressure is the same as atmospheric pressure. However, some aircraft contain pressurized regions, which are actively or passively pressurized relative to atmospheric pressure, and if the electrical conductor is located in a pressurized region, the ambient pressure will differ from the atmospheric pressure.

(16) The first electrical insulation **32** or second electrical insulation **34** can be sized to prevent partial discharge or insulation failure of the first electrical conductor **22** and the second electrical conductor **24** at the ambient air pressure, when the electrical power source **14** is at maximum voltage.

(17) By way of non-limiting example, the first electrical insulation **32** can be sized to prevent partial discharge or insulation failure of the first electrical conductor **22** at maximum voltage throughout the flight envelope of the aircraft propulsion system **12**. Optionally, the second electrical insulation **34** is sized to not prevent partial discharge or insulation failure of the second electrical conductor **24** at maximum voltage throughout the flight envelope of the aircraft propulsion system **12**.

(18) At least one sensor **40** can be used to obtain ambient air pressure or information related to the ambient air pressure. Ambient air pressure can be pressure at any point in or around the aircraft **10**. The at least one sensor **40** can be located inside the aircraft **10**, as shown. It is contemplated that the at least one sensor **40** can be coupled to any portion of the aircraft **10** including an exterior **42** of the aircraft **10**. It is further contemplated that the at least one sensor **40** can be multiple sensors mounted to any portion of the aircraft **10**. The at least one sensor **40**, by way of non-limiting example, can be one or more of a pressure sensor, an altimeter, a humidity sensor, a global positioning sensor (GPS), or a temperature sensor.

(19) The at least one sensor **40** can be operably coupled to the voltage regulator **18**. Optionally, a controller **44** can provide communication from the at least one sensor **40** to the voltage regulator **18**. Many known types of controllers can be used for the controller **44**. The specific type of controller is not germane to the present disclosure. The controller **44** can be included in or communicate with known flight management systems.

(20) The aircraft propulsion system **12** is illustrated, by way of non-limiting example, as coupled to or housed within a fuselage **46** of the aircraft **10**. It is contemplated that any portion of the aircraft propulsion system **12** can be coupled to or housed within other components of the aircraft **10** that can include, but are not limited to, wings **48**.

(21) While the voltage regulator is illustrated, by way of example, as a single voltage regulator **18**. Any number of voltage regulators can be used in the aircraft **10** to regulate the power to the electric motor **16**. It is contemplated that the voltage regulator **18** can regulate the voltage at the first conductor **22**, the second conductor **24**, or at both the first conductor **22** and the second conductor **24**. It is further contemplated that the voltage regulator **18** can regulate the voltage of any electrical conductor within the aircraft **10**.

(22) FIG. 2 schematically illustrates an aircraft **110** with an aircraft propulsion system **112** that is substantially similar to the aircraft **10** with the aircraft propulsion system **12**. Therefore, like parts will be identified with like numerals increased by 100, with it being understood that the description of the like parts of the aircraft propulsion system **12** applies to the aircraft propulsion system **112** unless otherwise noted.

(23) The aircraft propulsion system **112** includes an electric power source **114**, an electric machine **116**, a voltage regulator **118**, a first propeller **120a**, and a second propeller **120b**. The electric power source **114** includes a combustion engine **126** with a generator **128**, where the combustion engine

126 and the generator **128** can be formed together or separately. Optionally electrical communicator **130** with insulation **131** can provide electrical connection between the combustion engine **126** and the generator **128**. The insulation **131** can be similar to a first electrical insulation **132**. The first propeller **120a** is rotatably coupled to the combustion engine **126**.

(24) A first electrical conductor **122** with the first electrical insulation **132** can electrically connect the generator **128** of the electric power source **114** and the voltage regulator **118**. A second electrical conductor **124** with a second electrical insulation **134** can electrically connect the voltage regulator **118** to the electric machine **116**. Optionally, the second electrical conductor **124** can couple the voltage regulator **118** to an inverter **136**. An electrical communicator **138** with an insulation **139** can connect the inverter **136** to the electric machine **116**. The insulation **139** can be similar in size or insulating capability to the second electrical insulation **134**. The second propeller **120b** is operably rotated by the electric machine **116**.

(25) The electric machine **116** is powered by the electric power source **114** via the voltage regulator **118**. The voltage regulator **118** regulates the electrical power or voltage provided to the electric machine **116** from the electrical power source **114**.

(26) It is contemplated that the inverter **136** can be any combination of known electronics used to alter an electrical signal and could further include additional voltage control mechanisms.

(27) FIG. 3 schematically illustrates an aircraft **210** with an aircraft propulsion system **212** that is substantially similar to the aircraft **10** with the aircraft propulsion system **12**. Therefore, like parts will be identified with like numerals increased by 200, with it being understood that the description of the like parts of the aircraft propulsion system **12** applies to the aircraft propulsion system **212** unless otherwise noted.

(28) The aircraft propulsion system **212** includes an electric power source **214**, multiple electric machines **216**, a voltage regulator **218**, and multiple propellers **220**. The electric power source **214** includes a combustion engine **226** with a generator **228**, where the combustion engine **226** and the generator **228** can be formed together or separately. Optional electrical communicator **230** with an insulation **231** can provide electrical connection between the combustion engine **226** and the generator **228**. The insulation **231** can be similar to a first electrical insulation **232**.

(29) A first electrical conductor **222** with the first electrical insulation **232** can electrically connect the generator **228** of the electric power source **214** and the voltage regulator **218**. A second electrical conductor **224** with a second electrical insulation **234** can electrically connect the voltage regulator **218** to the electric machines **216**. Each of the multiple electric machines **216** then rotatable operates one of the multiple propellers **220**, as illustrated.

(30) Optionally, the second electrical conductor **224** can couple the voltage regulator **218** to multiple inverters **236**, where each of the multiple inverters **236** are electrically connected to one of the multiple electrical machines **216**. Electrical communicators **238** with insulation **239** can connect each of the multiple inverters **236** to one of the multiple electric machines **216**.

(31) The electric machines **216** are powered by the electric power source **214** via the voltage regulator **218**. The voltage regulator **218** regulates the electrical power provided to the electric machine **216** from the electrical power source **214**.

(32) It is contemplated that the voltage regulator **218** can be multiple voltage regulators, where each of the multiple voltage regulators couples to one of the multiple inverters **236** or the multiple electric machines **216**.

(33) It is further contemplated that the multiple inverters **236** can be any combination of electronics and could further include voltage control mechanisms.

(34) FIG. 4 schematically illustrates an aircraft **310** with an aircraft propulsion system **312** that is substantially similar to the aircraft **210** with the aircraft propulsion system **212**. Therefore, like parts will be identified with like numerals increased by 100, with it being understood that the description of the like parts of the aircraft propulsion system **212** applies to the aircraft propulsion system **312** unless otherwise noted.

(35) The aircraft propulsion system **312** includes an electric power source **314**, multiple electric machines **316**, a voltage regulator **318**, and multiple propellers **320**. The electric power source **314** includes an electrical storage machine **317**.

(36) A first electrical conductor **322** with the first electrical insulation **332** can electrically connect the electrical storage machine **317** of the electric power source **314** and the voltage regulator **318**. A second electrical conductor **324** with a second electrical insulation **334** can electrically connect the voltage regulator **318** to the electric machines **316**. Each of the multiple electric machines **316** then rotatable operates one of the multiple propellers **320**, as illustrated.

(37) Optionally, the second electrical conductor **324** can couple the voltage regulator **318** to multiple inverters **336**, where each of the multiple inverters **336** are electrically connected to one of the multiple electrical machines **316**. Electrical communicators **338** with insulation **339** can connect each of the multiple inverters **336** to one of the multiple electric machines **316**.

(38) The electric machines **316** are powered by the electric power source **314** via the voltage regulator **318**. The voltage regulator **318** regulates the electrical power provided to the electric machine **316** from the electrical power source **314**.

(39) It is contemplated that the voltage regulator **218** can be multiple voltage regulators, where each of the multiple voltage regulators couples to one of the multiple inverters **336** or the multiple electric machines **316**.

(40) It is further contemplated that the multiple inverters **336** can be any combination of electronics and could further include voltage control mechanisms.

(41) FIG. 5 illustrates a method **400** for operating an electric distribution system that includes the second electrical conductor or electrical conductor **24, 124, 224, 324** where the electrical conductor **24, 124, 224, 324** supplies electricity to the electric machine **16, 116, 216, 316** that rotationally drives the propeller **20, 120b, 220, 320** in an aircraft **10, 110, 210, 310**. At **202**, ambient air pressure of the electrical conductor **24, 124, 224, 324** is sensed. For example, the at least one sensor **40** can be used to obtain ambient air pressure such as atmospheric pressure of the electrical conductor **24, 124, 224, 324**. The sensor **40** can be located in a region of the aircraft **10, 110, 210, 310** known to have the same ambient air pressure or atmospheric pressure as the electrical conductor **24, 124, 224, 324**.

(42) At **404**, the voltage at the electrical conductor **24, 124, 224, 324** is controlled based on sensed ambient air pressure. For example, the at least one sensor **40** can be used to obtain ambient air pressure of the electrical conductor **24, 124, 224, 324**. The information can be communicated to the controller **44**. The controller **44** then determines a threshold voltage that corresponds to the ambient air pressure. The voltage regulator **18, 118, 218, 318** receives the threshold voltage from the controller **44**. The voltage at the electrical conductor **24, 124, 224, 324** is then controlled by the voltage regulator **18, 118, 218, 318** to be at or less than the threshold voltage. Controlling the voltage based on the sensed ambient air pressure can prevent partial discharge, accelerated aging, or other insulation failure of the electrical conductor **24, 124, 224, 324**.

(43) In operation, the aircraft propulsion system **12, 112, 212, 312** has a flight envelope defined by a region in which the aircraft **10, 110, 210, 310** with the aircraft propulsion system **12, 112, 212, 312** can operate safely based on a variety of factors that can include, but are not limited to, speed, altitude, air density, or angle of attack. The aircraft **10, 110, 210, 310** with the aircraft propulsion system **12, 112, 212, 312** includes the first electrical conductor **22, 122, 222, 322** with the first electrical insulation **32, 132, 232, 332** sized to prevent partial discharge or other insulation failure of the first electrical conductor **22, 122, 222, 322** at maximum voltage throughout the flight envelope of the aircraft propulsion system **12, 112, 212, 312**. The first electrical conductor **22, 122, 222, 322** electrically connects the electrical power source **14, 114, 214, 314** to the voltage regulator **18, 118, 218, 318**.

(44) During the flight envelope, the decrease in ambient air pressure indicates that there is less atmosphere to provide insulation to the first electrical conductor **22, 122, 222, 322**. The first

electrical insulator **32, 132, 232, 332** on the first electrical conductor **22, 122, 222, 322** must be thick enough to reduce the possibility of partial discharge, accelerated aging, or other insulation failure from the first electrical conductor **22, 122, 222, 322** at the highest altitude of the flight envelope when provided with maximum voltage.

(45) However, the aircraft **10** doesn't typically have the same power demands at higher altitudes as it does during take-off, for example. The reduced power demand at higher altitudes allows the aircraft propulsion system **12, 112, 212, 312** to utilize the voltage regulator **18, 118, 218, 318** to provide a reduced voltage or power to the electric machine **16, 116, 216, 316**. The second electrical conductor **24, 124, 224, 324** electrically connects the voltage regulator **18, 118, 218, 318** and the electric machine **16, 116, 216, 316**. Because the voltage regulator **18, 118, 218, 318** can control the voltage at the second electrical conductor **24, 124, 224, 324** to be less than or equal to at maximum voltage based on ambient air pressure, the second electrical insulation **34, 134, 234, 334** can be sized smaller than the first electrical conductor **22, 122, 222, 322**. Appropriate threshold voltage at the second electrical conductor **24, 124, 224, 324** can be determined using the at least one sensor **40** in communication with the controller **44**, resulting in the electric power or voltage being a function of the ambient air pressure. The threshold voltage at the second electrical conductor **24, 124, 224, 324** is determined based on the ambient air pressure of the second electrical conductor **24, 124, 224, 324** obtained by the at least one sensor. The threshold voltage prevents partial discharge of the second electrical conductor **24, 124, 224, 324**. Since, the electric machine **16, 116, 216, 316** requires less power at higher altitudes, especially when cruising, the operation of the aircraft **10, 110, 210, 310** is not impaired by the decrease in voltage at the second electrical conductor **24, 124, 224, 324**.

(46) Alternatively, the voltage regulator **18, 118, 218, 318** can provide constant power to the electric machine **16, 116, 216, 316** while adjusting the voltage to the threshold voltage as determined by the altitude. This is accomplished by allowing the current to vary accordingly to provide the constant power at varying threshold voltages.

(47) Technical effects of the system and method disclosed herein provides a control system that can vary the voltage of the supplied power as a function of the ambient air pressure of the electrical conductors. The ambient air pressure can, at least in part, be a function of altitude, making the voltage also a function of the altitude. By controlling the voltage at least some of the electrical conductors to require less electrical insulation to reduce the possibility of partial discharge from the electrical conductors.

(48) Additionally, controlling the voltage based on altitude can reduce the aging of the conductors. The manipulation of the voltage based on the threshold voltage determined by altitude can result in a lower cumulative current flowing through the conductors during a flight, therefore the conductors experience lower thermal impact. Similarly, the insulation experiences less wear with the decrease in heat due to current. A lower cumulative current flowing through the conductors can also result in a lower cumulative heat dissipation per flight.

(49) Aspects of the present disclosure provide for a variety of benefits. For example, by decreasing the size of the second electrical insulation, the weight of the aircraft will also decrease. A decrease in aircraft weight contributes to better fuel efficiency.

(50) Another advantage of the present disclosure is a cost savings due to the decreasing the size or amount of the second electrical insulation placed on the second electrical conductor.

(51) Many other possible configurations in addition to those shown in the above figures are contemplated by the present disclosure. For example, more than one aircraft propulsion system can be used on a single aircraft, where the multiple aircraft propulsion systems can include similar or different components. To the extent not already described, the different features and structures of the various aspects can be used in combination with others as desired. That one feature cannot be illustrated in all of the aspects is not meant to be construed that it cannot be, but is done for brevity of description. Thus, the various features of the different aspects can be mixed and matched as

desired to form new aspects, whether or not the new aspects are expressly described. Combinations or permutations of features described herein are covered by this disclosure.

(52) This written description uses examples to disclose aspects of the invention, including the best mode, and also to enable any person skilled in the art to practice aspects of the invention, including making and using any machines, devices, or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and can include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

(53) Further aspects of the invention are provided by the subject matter of the following clauses: 1. An aircraft propulsion system for an aircraft, including a propeller, an electric machine operably rotating the propeller, an electrical power source providing electrical power to the electric machine, and a voltage regulator regulating the electrical power provided to the electric machine from the electrical power source and configured to control the voltage of the electric power provided as a function of ambient air pressure in at least one portion of the aircraft. 2. The aircraft propulsion system of any preceding clause further including an electrical conductor connecting the voltage regulator to the electric machine and the at least one portion of the aircraft includes at least a portion of the electrical conductor. 3. The aircraft propulsion system of any preceding clause wherein the electrical conductor comprises an electrical insulation. 4. The aircraft propulsion system of any preceding clause wherein the electrical insulation is sized to prevent partial discharge at maximum voltage of the electrical power source at the ambient air pressure. 5. The aircraft propulsion system of any preceding clause wherein the electrical insulation is not sized to prevent partial discharge throughout the flight envelope of the aircraft propulsion system. 6. The aircraft propulsion system of any preceding clause wherein the voltage regulator is configured to control the voltage of the provided electric power to prevent partial discharge throughout the flight envelope. 7. The aircraft propulsion system of any preceding clause further comprising an ambient air pressure sensor operably coupled to the voltage regulator. 8. The aircraft propulsion system of any preceding clause wherein the ambient air pressure sensor is an altimeter. 9. The aircraft propulsion system of any preceding clause wherein an electrical conductor is exposed to atmospheric pressure and the ambient air pressure is the current atmospheric pressure. 10. The aircraft propulsion system of any preceding clause wherein the electric power source is one of an AC or DC electric power source. 11. The aircraft propulsion system of any preceding clause wherein the electric machine is at least one of an AC or DC motor. 12. The aircraft propulsion system of any preceding clause wherein the propeller comprises multiple propellers. 13. The aircraft propulsion system of any preceding clause wherein the electric machine comprises multiple electric machines, where each of the multiple propellers being rotatably driven by a different one of the multiple electric machines. 14. The aircraft propulsion system of any preceding clause where the voltage regulator is electrically coupled to all of the multiple electric machines. 15. The aircraft propulsion system of any preceding clause wherein the voltage regulator comprises multiple voltage regulators, with each of the multiple electric machines being electrically coupled to a different one of the multiple voltage regulators. 16. The aircraft propulsion system of any preceding clause wherein the electric power source comprises at least one of a combustion engine with generator or an electrical storage device. 17. The aircraft propulsion system of any preceding clause wherein the voltage regulator comprises an inverter or a converter. 18. A method of operating a electric distribution system having an electrical conductor supplying electricity to an electric machine rotationally driving a propeller in an aircraft, the method comprising sensing ambient air pressure of the electrical conductor and controlling the voltage at the electrical conductor based on the sensed ambient air pressure. 19. The method of any preceding clause wherein the controlling the voltage comprises controlling the voltage to prevent partial discharge from the electrical

conductor based on the sensed ambient air pressure. 20. The method of any preceding clause wherein the sensed ambient air pressure is atmospheric pressure.

Claims

1. An electric distribution system, comprising: an electrical power source having a power output; and a regulator configured to regulate the electrical power at the power output by controlling at least one of a voltage, a current, or a power provided to the electric distribution system based on an ambient air pressure.
2. The electric distribution system of claim 1, wherein the electric distribution system operably rotates an electric machine for a propeller.
3. The electric distribution system of claim 1, wherein the electric distribution system is an aircraft electric distribution system.
4. The electric distribution system of claim 3, wherein the ambient air pressure is sensed by an aircraft ambient air pressure sensor.
5. The electric distribution system of claim 1, further comprising electrical insulation insulating a conductor coupled at the power output, and wherein the electrical insulation is sized to prevent partial discharge at maximum voltage of the electrical power source at the ambient air pressure.
6. The electric distribution system of claim 5, wherein the voltage regulator is configured to regulate the power at the power output by controlling the voltage provided at the power output to prevent partial discharge.
7. The electric distribution system of claim 1, further comprising an electrical conductor connected at the power output, and wherein the electrical conductor is exposed to the ambient air pressure.
8. A method of operating an electric distribution system in an aircraft, the method comprising: sensing ambient air pressure; and controlling at least one of a voltage, a current, or a power of the electric distribution system based on the sensed ambient air pressure.
9. The method of claim 8 wherein sensing ambient air pressure includes sensing ambient air pressure at the electric distribution system.
10. The method of claim 8 wherein controlling the voltage includes controlling the voltage of a voltage regulator at a power output of an electrical power source of the electric distribution system.
11. The method of claim 8 wherein sensing ambient air pressure includes sensing an ambient air pressure of the aircraft.
12. The method of claim 8 wherein controlling the voltage includes controlling the voltage at a conductor of the electric distribution system, wherein the conductor is insulated by an insulator.
13. The method of claim 12 wherein controlling the voltage at the conductor further includes controlling the voltage to prevent partial discharge based on the sensed ambient air pressure.
14. The method of claim 8 wherein sensing ambient air pressure includes sensing atmospheric pressure.
15. A method of operating an electric distribution system in an aircraft, the method comprising: controlling at least one of a voltage, a current, or a power by a regulator, as a function of ambient air pressure in at least one portion of the aircraft; and providing, by the electric distribution system, the controlled at least one of the voltage, the current, or the power to an electric machine configured to operably rotate a propeller of the aircraft.
16. The method of claim 15, further comprising sensing ambient air pressure in the at least one portion of the aircraft.
17. The method of claim 16 wherein sensing ambient air pressure includes sensing atmospheric pressure.
18. The method of claim 15 wherein controlling the voltage includes controlling the voltage at a conductor of the electric distribution system, wherein the conductor is insulated by an insulator.
19. The method of claim 18 wherein controlling the voltage at the conductor further includes

controlling the voltage to prevent partial discharge based on the ambient air pressure.

20. The method of claim 15 wherein controlling the voltage includes controlling the voltage at a power output of an electrical power source of the electric distribution system.
